

Homework #3 - Solutions

1. Write the types of each variables in the below code segment according to their lifetimes and explain when memory is allocated and deallocated for each variable. (Hint: Assume that the programming language used for the below code segment is a new language which can support variable types in both C and JavaScript.)

```
int i , j;  
int main( ) {  
    int k, *p;  
    static double x;  
    ...  
    p = (int *)malloc(sizeof(int)*1000);  
    ...  
    y = "This is a string";  
    ...  
    y = 3.75;  
    ...  
    free(p);  
}
```

Answer: Types of variables according to their lifetimes are listed as follows:

i : static variable, memory is allocated before run-time, deallocated when the whole program finishes its execution.

j : static variable, memory is allocated before run-time, deallocated when the whole program finishes its execution.

k : stack-dynamic variable, memory is allocated during run-time when main function begins its execution, deallocated when the main function finishes its execution.

p : explicit heap-dynamic variable, memory is allocated during run-time when malloc function is called, deallocated when free function is called.

x : static variable, memory is allocated before run-time, deallocated when the whole program finishes its execution.

y : implicit heap-dynamic variable, memory is allocated during run-time when an assignment is made, deallocated when main function finishes its execution.

2. Consider the following skeletal C program:

```
void fun1(void); // prototype
void fun2(void); // prototype
void fun3(void); // prototype
```

```
void main(void){
    int x, y, z;
    // 1
    fun1( );
}
```

```
void fun1(void){
    int y, z, a;
    // 2
    fun2( );
}
```

```
void fun2(void){
    int z, a, b;
    // 3
    fun3( );
}
```

```
void fun3(void){
    int a, b, c;
    ...
    // 4
}
```

Write the referencing environments for the lines 1, 2, 3, and 4; if

- a) static scope is used
- b) dynamic scope is used.

Answer:

If static scope is used, referencing environment for the lines 1, 2, 3, and 4 are listed below:

Line #	Referencing Environment
1	x, y, z from main function
2	y, z, a from fun1 function
3	z, a, b from fun2 function
4	a, b, c from fun3 function

If dynamic scope is used, referencing environment for the lines 1, 2, 3, and 4 are listed below:

Line #	Referencing Environment
1	x, y, z from main function
2	x from main function y, z, a from fun1 function
3	x from main function y from fun1 function z, a, b from fun2 function
4	x from main function y from fun1 function z from fun2 function a, b, c from fun3 function

3. Assume the following rules of associativity and precedence for expressions:

Precedence	Highest	*, /, not +, -, &, mod Unary - ==, !=, <, <=, >, >=
	Lowest	and or, xor

Associativity Left-to-right

Show the order of evaluation of the following expressions by parenthesizing all subexpressions and placing a superscript or subscript on the right parenthesis to indicate order.

$a - b * 7 + \text{not } d$

$a * b == c + d \bmod 5 \text{ and } x != y$

$-a \text{ or } c == d \& f$

Answer:

$((a - (b * 7)_1)_3 + (\text{not } d)_2)_4$

$(((a * b)_1 == ((c + d)_2 \bmod 5)_3)_4 \text{ and } (x != y)_5)_6$

$((-a)_2 \text{ or } (c == (d \& f)_1)_3)_4$

4. Consider the following C program:

```
int fun(int *i){
    *i += 3;
    return 7;
}
void main(){
    int x = 5;
    x = x + fun(&x) + x + fun(&x);
}
```

What is the value of x after the assignment statement in main, assuming

- a) operands are evaluated left to right.
- b) operands are evaluated right to left.

Answer:

If operands are evaluated from left to right, the value of x is computed as follows:

$x = 5 + 7 + 8 + 7$ so, x will be equal to 27

If operands are evaluated from right to left, the value of x is computed as follows:

$x = 11 + 7 + 8 + 7$ so, x will be equal to 33